

Elementary Problems

Congruence

Version 1.0.1

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Basic Work

1. Prove that $n^3 \equiv n \pmod{6}$ for any integer n .
2. Solve the congruence equation $3n \equiv 2 \pmod{7}$.
3. Solve the congruence equation $4n \equiv 1 \pmod{6}$.
4. Solve the congruence equation $15n \equiv 5 \pmod{10}$.
5. Find all prime numbers p such that $p + 2$ and $p + 4$ are prime numbers.
6. Find all positive integers n such that $3n - 4$, $4n - 5$ and $5n - 3$ are prime numbers.
7. Find all positive integers n such that $2^n - 1$ and $2^n + 1$ are prime numbers.
8. Prove that the remainder when a positive integer n is divided by 9 is equal to that when the sum of digits of n is divided by 9.
9. (IMO 1975 Problem 4) When 4444^{4444} is written in decimal notation, the sum of its digits is A . Let B be the sum of the digits of A . Find the sum of the digits of B . (A and B are written in decimal notation.)
10. (IMO shortlist 1990) For a given positive integer k denote the square of the sum of its digits by $f_1(k)$ and let $f_{n+1}(k) = f_1(f_n(k))$. Determine the value of $f_{1991}(2^{1990})$.
11. (IMO shortlist 1967) Does there exist an integer such that its cube is equal to $3n^2 + 3n + 7$, where n is an integer?
12. (IMO shortlist 1969) Prove that the equation $\sqrt{x^3 + y^3 + z^3} = 1969$ has no integral solutions.
13. (USAMO 1979 Problem 1) Determine all nonnegative integral solutions $(n_1, n_2, \dots, n_{14})$ if any, apart from permutations, of the Diophantine equation

$$n_1^4 + n_2^4 + \dots + n_{14}^4 = 1599.$$

14. Find all integers m and n such that $m^3 + n^4 = 7$.
15. Find all integers m and n such that $m^2 + 4 = n^5$.
16. Let a and d be positive integers and p be a prime number. If none of the numbers

$$a + d, a + 2d, a + 3d, \dots, a + pd$$

is divisible by p , prove that $p \mid d$ and $p \nmid a$.

17. (All-Russian MO 1963) An infinite arithmetic progression of positive integers contains a square number. Prove that it contains infinitely many squares.
18. Let p be a prime number. The Fibonacci sequence $(F_n)_{n \geq 1}$ is defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$. Prove that the sequence contains infinitely many terms which are multiples of p .
19. (USAMO 1973 Problem 2) Let $\{X_n\}$ and $\{Y_n\}$ denote two sequences of integers defined as follows:

$$\begin{aligned} X_0 &= 1, X_1 = 1, X_{n+1} = X_n + 2X_{n-1} & (n = 1, 2, 3, \dots), \\ Y_0 &= 1, Y_1 = 7, Y_{n+1} = 2Y_n + 3Y_{n-1} & (n = 1, 2, 3, \dots). \end{aligned}$$

Prove that, except for the '1', there is no term which occurs in both sequences.

20. The sequence $(a_n)_{n \geq 1}$ satisfies $a_{n+1} = a_n^3 + 11$ for $n \geq 1$. Prove that the sequence contains at most one perfect square.
21. Let $P(n)$ be a polynomial with integer coefficients such that $P(n)$ is a prime for all positive integers n . Prove that $P(n)$ is a constant polynomial.
22. Find all integers $n \geq 1$ such that there exist complete residue systems $\{a_1, a_2, \dots, a_n\}$ and $\{b_1, b_2, \dots, b_n\}$ modulo n such that $\{a_1 + b_1, a_2 + b_2, \dots, a_n + b_n\}$ is a complete residue system.
23. (IMO 1985 Problem 2) Let n and k be given relatively prime natural numbers, $k < n$. Each number in the set $M = \{1, 2, \dots, n-1\}$ is coloured either blue or white. It is given that
 - (i) for each $i \in M$, both i and $n-i$ have the same colour;
 - (ii) for each $i \in M$, $i \neq k$, both i and $|i-k|$ have the same colour.

Prove that all numbers in M must have the same colour.

24. Prove or disprove the following.
 - (a) Let a, b and n be positive integers. If $ab \equiv 0 \pmod{n}$, then we have $a \equiv 0 \pmod{n}$ or $b \equiv 0 \pmod{n}$.
 - (b) Let a, b and n be positive integers. If $a^2 \equiv b^2 \pmod{n}$, then $a \equiv \pm b \pmod{n}$.
 - (c) Let a, b, c, d and m, n be positive integers. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{n}$, then $ac \equiv bd \pmod{mn}$.
 - (d) Let m and n be relatively prime positive integers. Then for any integer a , the equation $mnx \equiv a \pmod{m+n}$ has a unique solution $x \pmod{m+n}$.

- (e) Let n be a positive integer. Let $P(x)$ be a polynomial with integer coefficients of degree $d \geq 1$ such that the leading coefficient is relatively prime to n . Then the equation $P(m) \equiv 0 \pmod{n}$ has at most d solutions modulo n .

Chinese Remainder Theorem

25. Find all integers n such that

$$\begin{aligned} n &\equiv 2 \pmod{3}, \\ n &\equiv 5 \pmod{7}. \end{aligned}$$

26. Find all integers n such that

$$\begin{aligned} 3n &\equiv 1 \pmod{5}, \\ 2n &\equiv 4 \pmod{8}. \end{aligned}$$

27. Find all integers n such that

$$\begin{aligned} 2n &\equiv 1 \pmod{3}, \\ 3n &\equiv 1 \pmod{4}, \\ 2n &\equiv 3 \pmod{5}. \end{aligned}$$

28. Find all positive integers n such that the last three digits of n^2 are 144 in that order.

29. (IMO 1989 Problem 5) Prove that for each positive integer n there exist n consecutive positive integers none of which is an integral power of a prime number.

30. Prove that there are infinitely many even positive integers k such that $p^2 + k$ is composite for all prime numbers p .

31. (IMO shortlist 1987)

- (a) Let $\gcd(m, k) = 1$. Prove that there exist integers $a_1, a_2, \dots, a_m$ and $b_1, b_2, \dots, b_k$ such that each product $a_i b_j$ ($i = 1, 2, \dots, m; j = 1, 2, \dots, k$) gives a different residue when divided by mk .

32. (CWMI 2012 Problem 1) Find the smallest positive integer m such that for any prime number p larger than 3, we have

$$105 \mid 9^{p^2} - 29^p + m.$$

33. (IMO shortlist 1966) We take 100 consecutive natural numbers $a_1, a_2, \dots, a_{100}$. Determine the last two digits of the number $a_1^8 + a_2^8 + \dots + a_{100}^8$.

Euler-Fermat Theorem

34. Find the remainder when 2^{1000} is divided by 17.
35. Find the last two digits of 3^{1234} .
36. Find the remainder when 5^{50} is divided by 221.
37. (IMO 1971 Problem 3) Prove that the set of integers of the form $2^k - 3$ ($k = 2, 3, \dots$) contains an infinite subset in which every two members are relatively prime.
38. Find all prime numbers p such that $11 \mid 2^p + 3^p$.
39. Prove that there does not exist a prime number p such that p divides $2^{p-1} + 3^{p-1}$.
40. Find all prime numbers p and q such that $pq \mid p^q + q^p + 7$.
41. Find all integers m and n such that $m^3 - n^3 = 2013$.
42. (IMO shortlist 1993 N2) A natural number n is said to have the property P , if, for all a , n^2 divides $a^n - 1$ whenever n divides $a^n - 1$.
 - (a) Show that every prime number n has property P .
 - (b) Show that there are infinitely many composite numbers n that possess property P .
43. (Canada MO 1983 Problem 4) Prove that for every prime number p , there are infinitely many positive integers n such that p divides $2^n - n$.
44. Let m and n be relatively prime positive integers. Prove that there exists a positive integer k such that $m^k + n^k - 1$ is divisible by mn .
45. (CWMO 2008 Problem 3) Let $m \geq 2$ be an integer, and let $a_1, a_2, \dots, a_m$ be positive integers. Prove that there exist infinitely many positive integers n such that the number $a_1 \cdot 1^n + a_2 \cdot 2^n + \dots + a_m \cdot m^n$ is a composite number.
46. Let k be a given positive integer and p be a prime number larger than $k + 1$. Prove that there exists a positive integer n such that all of the numbers

$$n^n, (n+1)^n + 1, (n+2)^n + 2, \dots, (n+k)^n + k$$

are not divisible by p .

47. (Baltic Way 2005) Let n be a positive integer, let p be prime and let q be a divisor of $(n+1)^p - n^p$. Show that p divides $q - 1$.

48. (CGMO 2010 Problem 3) Prove that for each positive integer n , there exist a prime number p and an integer m satisfying the following three conditions:

- (1) $p \equiv 5 \pmod{6}$;
- (2) $p \nmid n$;
- (3) $n \equiv m^3 \pmod{p}$.

Record

v1.0: problems 1–48